

JAMES RUSE AGRICULTURAL HIGH SCHOOL
YEAR 11 MATHEMATICS
HALF YEARLY EXAM 2005

QUESTION 1

	Marks
(a) Graph on the number line $\{x: -1 \leq x \leq 3\}$	1
(b) Simplify $(2^x)^x$	1
(c) Solve $ 5x - 6 \leq 12$	2
(d) Shade the region $(x - 1)^2 + (y - 1)^2 \leq 1$	1
(e) Find to 2 decimal places the solution of $3^x = 7$	2
(f) Rationalize the denominator : $\frac{3\sqrt{2} - 5}{4 + \sqrt{2}}$	2
(g) Solve $x^2 \geq 121$	2
(h) Solve $2 \sin x + 1 = 0$ for $\{-90^\circ \leq x \leq 90^\circ\}$	1
(i) Graph $y = x(x + 1)(x + 3)^2$, hence solve $x(x + 1)(x + 3)^2 \leq 0$	3

QUESTION 2 (START A NEW PAGE)

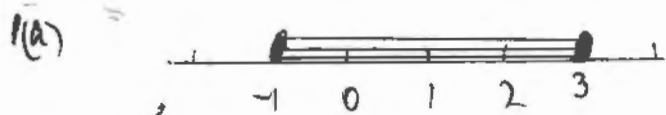
(a) Triangle ABC is represented by the points $A(3, -2)$, $B(4, 5)$ and $C(-6, 1)$. Find : (i) the equation of line AB in general form. (ii) the equation of the line perpendicular to AB passing through point C . (iii) the length of the altitude from A to BC if the equation of BC is given by $2x - 5y + 17 = 0$. (iv) the co-ordinates M of the midpoint AB . (v) the length of the median from C to AB . (vi) the co-ordinates of point D if $ABCD$ is a parallelogram.	2 2 2 1 2 2
(b) The roots of $2x^2 - 3x - 5 = 0$ are α and β . Find the value of : (i) $\alpha + \beta$ (ii) $\alpha\beta$ (iii) $\alpha^2 + \beta^2$	 1 1 2

QUESTION 3 (START A NEW PAGE)

- (a) Differentiate with respect to x :
- (i) $\frac{9x^2 - 11}{x}$ 2
- (ii) $\sqrt{x}(3x - 5)$ 2
- (b) Find the equation of the tangent to the curve $y = 4x^3 - 8x + 7$ at $x = -1$. 3
- (c) Graph $y = \tan x^\circ$ in the domain $\{0^\circ \leq x^\circ \leq 180^\circ\}$ 2
- (d) Find the value of x giving reasons $(6 - x)(x + 2) = 3x$ 2
- (e) A plane travels 250 km on a bearing $N57^\circ E$, then 380 km on a bearing of $193^\circ T$.
- (i) How far (to nearest km) is the plane from the initial position? 2
- (ii) What is the final bearing of the plane from the initial position ? 2

QUESTION 4 (START A NEW PAGE)

- (a) Graph $y = \sqrt{16 - x^2}$. 1
- State the domain and range. 2
- (b) Find the Cartesian equation of the locus of point $P(x, y)$ if point P is equidistant from $A(3, -2)$ and $B(-1, 4)$. 2
- (c) Find the values of k such that $(2k + 1)x^2 - 5x + 3$ is positive definite. 2
- (d)(i) Triangle OTS is right angled at S and ORS is a sector with OS and OR equal to one unit. T 2
- Prove the inequality : R
- $\sin x < x < \tan x$ for $\{ 0 < x < \frac{\pi}{2} \}$
- (ii) O x S 1
- Hence deduce the value of $\lim_{x \rightarrow 0} \frac{x}{\sin x}$. 2
- (e) A circular disc 8cm in diameter touches a flat surface at the point P .
- If the disc rolls 35 cm how high is the point P above the flat surface ?
- (f) Solve for n : $2^{2n+1} - 9(2^n) + 4 = 0$ 3

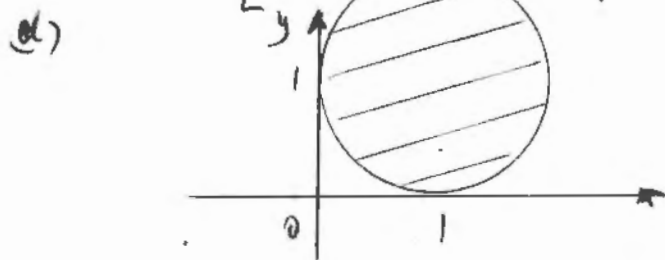


(b) x^2

(c) $5x-6 = \pm 12$

$5x = 6 \text{ or } 18$
 $x = -1\frac{1}{5} \text{ or } 3\frac{3}{5}$

Soln $\left\{ -1\frac{1}{5} \leq x \leq 3\frac{3}{5} \right\}$

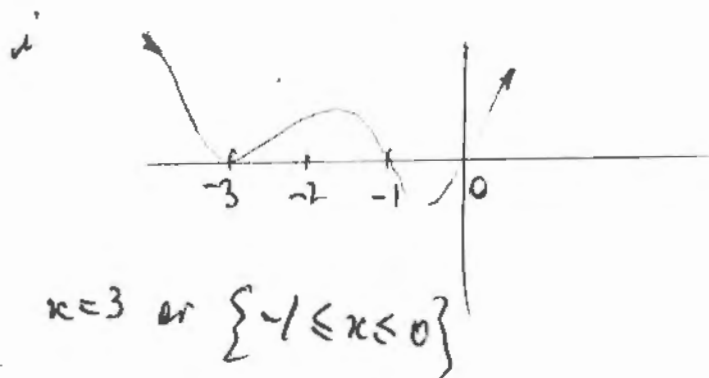


(e) $3^x = 7$
 $x = \log_3 7$
 $= \frac{\log 7}{\log 3}$
 $= 1.77$

(f) $\frac{(3\sqrt{2}-5)(4-\sqrt{2})}{(4+\sqrt{2})(4-\sqrt{2})}$
 $= \frac{12\sqrt{2}-6-20+5\sqrt{2}}{16-2}$
 $= \frac{17\sqrt{2}-26}{18}$

(g) $\{x \geq 4\}$ or $\{x \leq -4\}$

(h) $\sin x = -\frac{1}{2}$
 $x = -30^\circ$



2a(i) $m_{AB} = 7$
 $y+2 = 7(x-3)$

$7x-y-23=0$

(ii) $x+7y = -6+7$

$x+7y-1=0$

(iii) $D = \frac{6+10+17}{\sqrt{2^2+5^2}}$

$= \frac{33}{\sqrt{29}}$ units.

(iv) $M\left(\frac{7}{2}, \frac{3}{2}\right)$

(v) $D = \sqrt{\left(\frac{21}{2}\right)^2 + \left(-\frac{1}{2}\right)^2}$
 $= \frac{\sqrt{442}}{2}$ units

(vi) $M_{AC} = \left(-\frac{3}{2}, -\frac{1}{2}\right)$

$M_{BD} = \left(\frac{x+4}{2}, \frac{y+5}{2}\right)$

$D \subseteq (-7, -6)$

(b) (i) $2+p = \frac{3}{2}$

(ii) $2p = -\frac{5}{2}$

(iii) $2^2+p^2 = (2+p)^2 - 2p$
 $= \left(\frac{3}{2}\right)^2 - 2 \cdot -\frac{5}{2}$
 $= 7\frac{1}{4}$

3(b) (i) $9 + \frac{11}{x^2}$

(ii) $\frac{d}{dx} \left[3x^{\frac{3}{2}} - 5x^{\frac{1}{2}} \right] = \frac{9}{2} x^{\frac{1}{2}} - \frac{5}{2} x^{-\frac{1}{2}}$

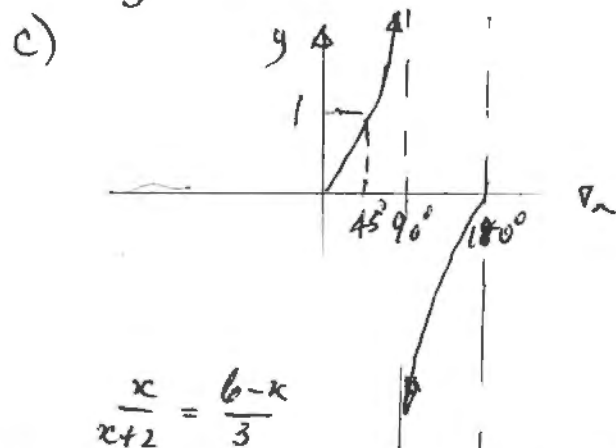
b) $y = 4x^3 - 8x + 7$

$\frac{dy}{dx} = 12x^2 - 8$

$m = 12 - 8 = 4$

$x = -1$
 $y = 11$

Eqn Tangent
 $y - 11 = 4(x + 1)$
 $4x - y + 15 = 0$

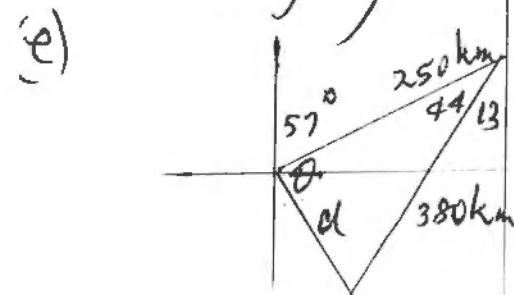


$\frac{x}{x+2} = \frac{6-x}{3}$

d) $(6-x)(x+2) = 3x$
 $12 + 4x - x^2 = 3x$
 $x^2 - x - 12 = 0$
 $(x-4)(x+3) = 0$
 $x = 4 \text{ or } x = -3$
But $\{0 < x < 6\}$

(A line parallel to one side of a triangle divides the other two in the same ratio)

$x = 4$ units only



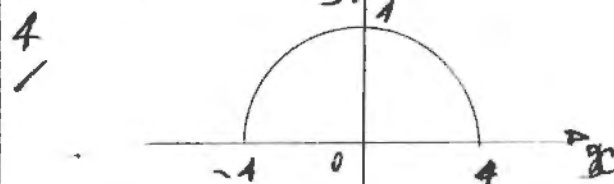
$d^2 = 250^2 + 380^2 - 2 \times 250 \times 380 \cos 57^\circ$
 $= 70225$
 $d = 265 \text{ km}$

(ii) $380^2 = 70225 + 250^2 - 2 \times 250 \times 265 \cos \theta$

$\cos \theta = -0.04$

$\theta = 95^\circ 3'$

Brg 1520°



Domain $\{-4 \leq x \leq 4\}$

Range $\{0 \leq y \leq 4\}$

b) $(x-3)^2 + (y+2)^2 = (x+1)^2 + (y-4)^2$
 $x^2 - 6x + 9 + y^2 + 4y + 4 = x^2 + 2x + 1 + y^2 - 8y + 16$
 $-8x + 12y - 4 = 0$
 $2x - 3y + 1 = 0$

c) $2k + 170$
 $k > -\frac{1}{2}$

And $\Delta < 0$

$25 - 4 \times 3 \times (2k+1) < 0$

$25 - 24k - 12 < 0$

$-24k < -13$

$k > \frac{13}{24}$

$\therefore$ Soln $k > \frac{13}{24}$

d(i) Area $\Delta ORS < \text{Area Sector} < \text{Area } \Delta ORT$

$\frac{1}{2} \cdot 1 \cdot 1 \sin x < \frac{1}{2} \cdot 1^2 \cdot x < \frac{1}{2} \cdot 1 \cdot \tan x$

$\sin x < x < \tan x$

(ii) $1 < \frac{x}{\sin x} < \frac{\tan x}{\cos x}$

$\lim_{x \rightarrow 0} (1) < \lim_{x \rightarrow 0} \left(\frac{x}{\sin x} \right) < \lim_{x \rightarrow 0} \left(\frac{1}{\cos x} \right)$

$1 < \lim_{x \rightarrow 0} \frac{x}{\sin x} < 1$

$\therefore \lim_{x \rightarrow 0} \frac{x}{\sin x} = 1$

$$(e) \quad \theta = +0$$

$$35 = \cancel{40}$$

$$\theta = 8.75 \text{ rad.}$$

$$\begin{aligned} \text{height} &= 4 + 4 \sin\left(8.75 - \frac{\pi}{2}\right) \\ &= 7.12 \text{ cm.} \end{aligned}$$

$$(f) \quad 2^{2n+1} - 9(2^n) + 4 = 0$$

$$2 \cdot (2^n)^2 - 9(2^n) + 4 = 0$$

$$\text{let } u = 2^n$$

$$2u^2 - 9u + 4 = 0$$

$$(2u-1)(u-4) = 0$$

$$u = \frac{1}{2} \quad \text{or} \quad u = 4$$

$$2^n = 2^{-1} \quad \text{or} \quad 2^n = 2^2$$

$$n = -1 \quad \text{or} \quad n = 2$$